INCH-POUND

MIL-DTL-85547C (OS)
11 January 2013
SUPERSEDING
MIL-I-85547B (AS)
30 November 1990

DETAIL SPECIFICATION

IGNITER, PVU-3A/E

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and inspection of the Igniter, PVU-3A/E, herein after referred to as the igniter.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-R-85545	Rocket, GTR-18A

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810	Environmental Engineering Considerations And Laboratory Tests
MIL-STD-1168	Ammunition Lot Numbering And Ammunition Data Card

Comments, suggestions, or questions to this document should be addressed to DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E12, Standardization, 3703 Benson Road, Indian Head, Maryland 20640-5085, OFFICIAL BUSINESS, or emailed to IHDIV STANDARDIZATION TEAM. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1370
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:03Z
Check the source to verify that this is the current version before use.

DOD-STD-2101 Classification of Characteristics

(Copies of these documents are available online at https://assist.dla.mil or from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

NAVAL AIR SYSTEMS COMMANO (NAVAIR) (CAGE Code 30003)

DL1335AS275	Data List, PVU-3A/E
1335AS275	Igniter, PVU-3A/E
1335AS280	Rocket, GTR-18A
1335AS290	Packaging Assembly
1335AS390	Simulator, Surface-to-Air Missile SMU-124/E (Smokey SAM Simulator)

(Unless otherwise indicated, copies of Naval Air Systems Command drawings are available from the Naval Air Technical Services Facility (NATSF) (Code 312), 700 Robbins Avenue, Philadelphia, PA 19111-5096.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

American Society For Quality (ASQ)

ANSI/ASQ Z1.4 Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at http://e-standards.asq.org/ or from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203)

American Society For Testing And Materials (ASTM)

ASTM-E1742/E1742M Standard Practice for Radiographic Examination

(Copies of these documents are available online at http://www.astm.org or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:03Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Design and construction. The igniter shall be fabricated and assembled in accordance with the following and all documents listed thereon:

PVU-3A/E DL1335AS275

For the Smokey SAM Simulator, utilizing the PVU-3A/E igniter, see Drawing 1335AS390.

3.3 Performance. The igniter shall meet the following requirements where initiated by a firing pin (see 4.5.6):

a. (M101) The igniter shall not exhibit ignition failure (see 6.6.1).
b. (M102) The igniter tube shall not rupture.
c. (M103) The igniter shall not exhibit burn-through of the cartridge case.
d. (M104) The cartridge case shall not separate from the igniter tube.

3.4 Environmental conditioning. The igniter shall meet the performance requirements of 3.3 after exposure to vibration and shock as specified in 4.5.4 and 4.5.5.

3.5 Radiographic. The igniter shall not exhibit any missing or extra components. The configuration and orientation of all PVU-3A/E components shall be in accordance with Drawing 1335AS275 (see 4.5.1).

3.6 Marking. Unless otherwise specified in the contract (see 6.2), PVU-3A/E igniter markings shall be in accordance with Drawing 1335AS275.

3.7 Data cards. Data cards shall be prepared in accordance with MIL-STD-1168 and shall be furnished for each igniter lot (see 4.4.1).

3.8 Workmanship. All parts shall be free from chips, dirt, grease, rust and all other foreign materials. The igniter shall be free from explosive material on all external surfaces. There shall be no burrs or other imperfections of manufacturing which would adversely affect the assembly, functioning, or handling of the igniter. The igniter shall not exhibit defective bonds or poor molding fabrication.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified (see 4.6 and 6.2), all inspections shall be performed under the following conditions:

Temperature 73°F ± 18°F
Relative humidity 50 ± 30 percent
Atmospheric pressure 14.0 + 1.0, - 1.5 psia

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:03Z
Check the source to verify that this is the current version before use.

4.3 First article inspection. First article inspection shall be performed on sample units which have been produced using the same materials equipment, processes, and procedures normally used in production.

4.3.1 Sample. First article sample shall consist of 36 igniters (see 6.4).

4.3.2 Inspection routine. The igniters shall be divided into three equal groups and subjected to the tests specified in figure 1. The sequence of tests shall be as shown in figure 1.

4.3.3 Failure. Failure of any igniter shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the tests in 4.5.1, 4.5.2 and 4.5.6 in sequence.

4.4.1 Inspection lot. Igniter inspection lot definition, formation, and size shall be in accordance with ANSI/ASQ Z1.4. In addition, an inspection lot of igniters shall consist of assemblies meeting the following conditions:

a. All primers in an igniter inspection lot shall come from a single primer inspection lot

b. All ignition grains in an Igniter lot shall come from a single ignition-grain inspection lot.

c. Lot size shall be not greater than 20,000 igniters

4.4.2 Lot numbering. Unless otherwise specified (see 6.2), lot numbering shall be in accordance with MIL-STD-1168. Lot interfix numbers shall be assigned by the acquiring activity.

4.4.3 Sampling plan. Sampling for visual, radiographic and first article static-fire inspections shall be in accordance with ANSI/ASQ Z1.4, inspection level 1.

4.4.4 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Requirements that are critical are identified by the symbol (C) and those that are major are identified by the symbol (M). The number following the classification symbol indicates the serial number of the requirement. Requirements not annotated with a classification code are classified minor.

4.5 Methods of inspection.

4.5.1 Radiographic. The igniter shall be radiographically inspected perpendicular to its longitudinal axis in accordance with ASTM E 1742.

4.5.2 Visual examination. Igniters shall be visually examined for conformance to the requirements of 3.2 and 3.8. Four power (4X) magnification may be used for this inspection.

4.5.3 Detail strip. The igniters shall be disassembled and each component examined for conformance to the requirements of 3.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:03Z
Check the source to verify that this is the current version before use.

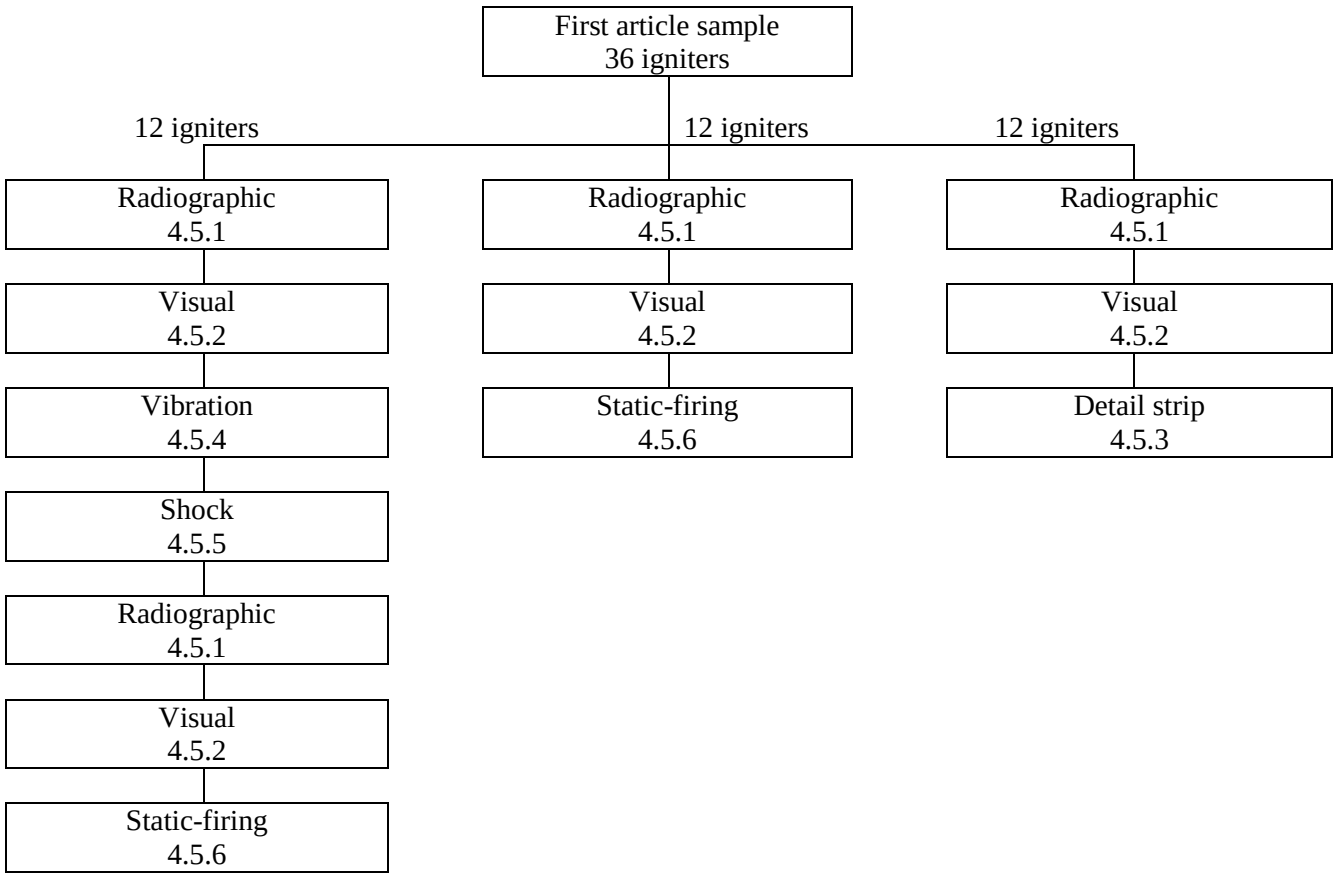


FIGURE 1. First article inspection.

4.5.4 Vibration. The igniters shall be preserved in plastic bags in accordance with 5.1. Two bags of igniters shall be packed in the Smokey SAM shipping container in accordance with Drawing 1335AS290. The igniters shall be subjected to a vibration test with 12 GTR-18A rockets (see MIL-R-85545 and Drawing 1335AS280) (or 12 dummy rockets that simulate the GTR-18A weight configuration approved by the acquiring activity) in the shipping container. The shipping container shall be banded with strapping in accordance with Drawing 1335AS290. While in the shipping configuration, the igniters shall be subjected to the vibration test of MIL-STD-810, method 514.5, category 4 per table 514.5-I. Vibration test levels given in figures 514.5C-1 through 514.5C-3 shall be used. Test time shall be based on a transportation of 3000 miles.

4.5.5 Shock. The igniters shall be removed from their bags and shipping containers for this test. The 6-foot drop test shall be conducted in accordance with MIL-STD-810, method 516.5, procedure IV. Drop height shall be measured from the lowest point of the test item to the impact plane. One-third of the igniters shall be dropped with the primer end down, one-third with the primer end up, and one-third with the longitudinal axis parallel to the impact plane. Drop orientation of the igniters shall be required at the drop point only.

4.5.6 Static-firing. Static-firing shall be performed at ambient conditions. A test fixture, equipped with a firing pin capable of producing an indentation not less than 0.015 inch without rupture in the igniter primer, shall be used to perform this test. The igniter shall be inserted into the test fixture in a vertical position, cartridge down. The igniter shall be fired by hitting the percussion primer. Ballistic test

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:03Z
Check the source to verify that this is the current version before use.

requirements of MIL-R-85545 using the associated sampling plan may be used to satisfy quality conformance static fire requirement for a quantity of igniters being packaged with a single lot of GTR-18A rockets. Following the static firing, verify that all performance requirements have been met (see 3.3).

4.6 First article and quality conformance inspection report. A first article or quality conformance report (see 6.3) shall be available to the acquiring activity within the time frame specified (see 6.2). The report shall contain the following information:

a. Date of report

b. Date of inspection

c. Lot number

d. Manufacturer

e. Operational objectives

f. Test equipment used

g. Environmental conditions of inspection

h. Type of failure, if any

i. Analysis, significance, and recommendations

j. Disposition of parts

k. Corrective action taken

4.7 Inspection of Packaging. The sampling and inspection of the preservation, packing, and container marking shall conform to this specification (see section 5).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2, and 6.6). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:03Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or. explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The igniter is intended for use with the GTR-18A rocket (see drawing 1335AS280) and the LMU-23A/E (see drawing 1335AS380) or LMU-24A/E (see drawing 1335AS700) launcher to train aviation and ground forces to recognize and cope with attacking surface-launch missiles. Since the igniter was developed strictly for use with these military ordnance items, there is no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. If required, the specific issue of individual documents referenced (see 2.2).
c. Part number.
d. Quantity required.
e. Whether first article inspection is required (see 6.4), and if so, specify the test activity.
f. Date and place of delivery.
g. Inspection conditions, if other than specified in 4.2.
h. Preservation, packing, marking and lot numbering instructions, if other than specified in 3.6, 4.4.2, and 6.5.
i. Time frame for submission of first article or quality conformance
j. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" OP5 are applicable.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The specific acquisition should be reviewed to ensure that only essential data are requested/provided. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	Title
4.6	Report, test

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 36 igniters. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval its presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:03Z
Check the source to verify that this is the current version before use.

6.5. Packaging.

6.5.1 Preservation. Unless otherwise specified in the contract (see 6.2), igniters should be preserved in 4.5 inch x 14 inch bags in accordance with MIL-DTL-117, type I, class B, style 2. Nominal thickness of the bags should be a minimum of 0.003 inch. Six igniters should be placed in each bag with the primer end at the bottom. The top of the bags should then be heat-sealed.

6.5.2 Packing. Unless otherwise specified in the contract (see 6.2), bagged igniters should be packed in accordance with Class C explosive regulations of 49 CFR 171-178 and industrial practice. Interior bracing for the igniters should afford protection against physical damage or deformation to the igniters and the packaging material. The weight of the container contents should not be greater than 60 pounds.

6.5.3 Marking. Unless otherwise specified in the contract (see 6.2), marking on the shipping container should be in accordance with 49 CFR 171-178 and MIL-STD-129.

6.6 Definition. For the purpose of this specification, the following definition applies.

6.6.1 Ignition failure. Ignition failure is the failure of the igniter grain to burn after application of sufficient firing pin energy to produce an indentation of the primer of not less than 0.015 inch.

6.7 Subject term (key word) listing.

Cartridge case
Firing pin energy
Ignition failure
Ignition grains
Static firing

6.8 Changes from Previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes. Changes to this specification were authorized by ECP IH06020.

Custodian:
Navy – OS

Preparing activity:
Navy - OS
(Project No. 1370-2012-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:03Z
Check the source to verify that this is the current version before use.